

APEXPLANET CYBERSECURITY LAB REPORT

Title: Cybersecurity Lab Setup Using Kali Linux and Metasploitable2

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1. OBJECTIVE

The objective of this lab is to set up a cybersecurity testing environment using Kali Linux and Metasploitable2 in VirtualBox. The lab demonstrates virtual machine setup, network communication, packet capture, port scanning, and cryptography using industry-standard cybersecurity tools.

2. TOOLS USED

- Oracle VirtualBox
- Kali Linux
- Metasploitable2
- Wireshark
- Nmap
- OpenSSL

3. INTRODUCTION

Cybersecurity professionals use isolated lab environments to safely learn and practice security concepts. Kali Linux is a popular penetration testing operating system that contains numerous security tools. Metasploitable2 is an intentionally vulnerable Linux machine designed for security testing and training. In this lab, both virtual machines were configured and connected using a Host-Only Network in VirtualBox.

4. LAB SETUP

Step 1:

Oracle VirtualBox was installed on the host machine.

Step 2:

Kali Linux virtual machine was imported and configured successfully.

Step 3:

Metasploitable2 virtual machine was imported and configured successfully.

Step 4:

Both virtual machines were connected using a Host-Only Adapter to enable communication between them.

Step 5:

The network configuration was verified using IP address commands.

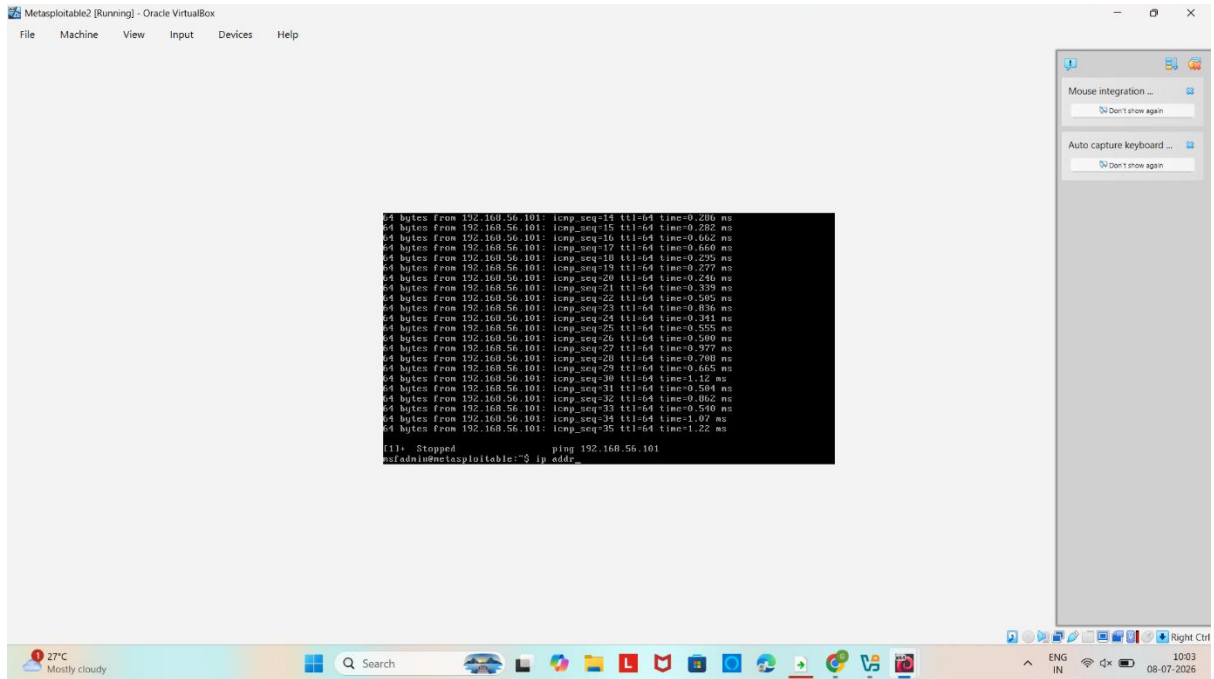
Kali Linux IP Address:

192.168.56.101

Metasploitable2 IP Address:

192.168.56.102





5. NETWORK CONNECTIVITY TEST

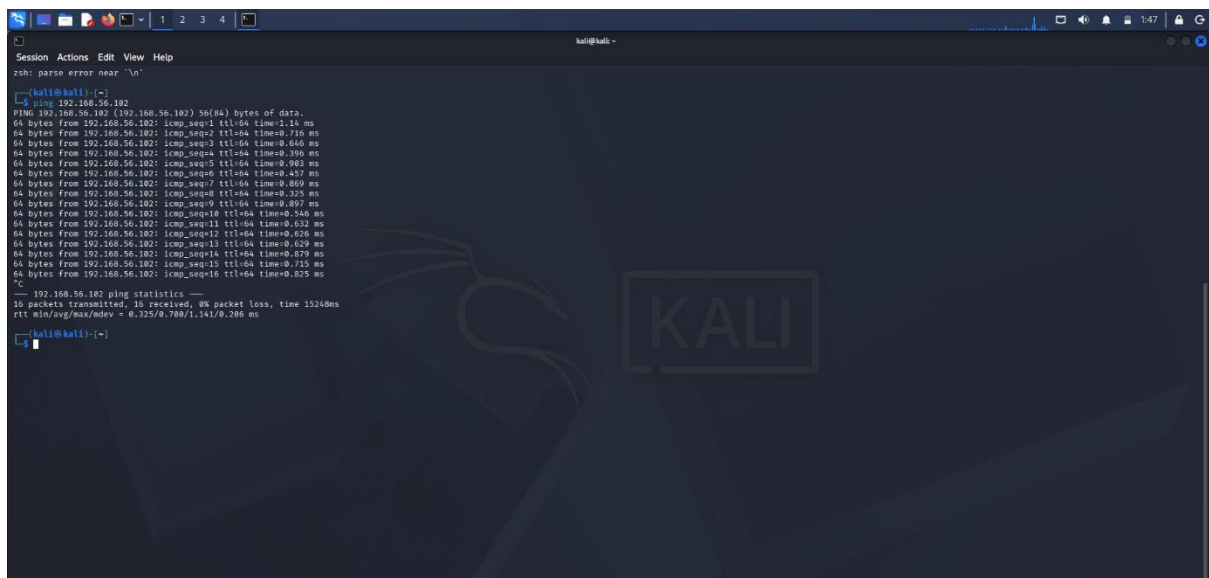
A connectivity test was performed using the ping command to verify communication between Kali Linux and Metasploit2.

Command Used:

ping 192.168.56.102

Result:

Successful replies were received from the target machine, confirming proper network communication.



6. PACKET ANALYSIS USING WIRESHARK

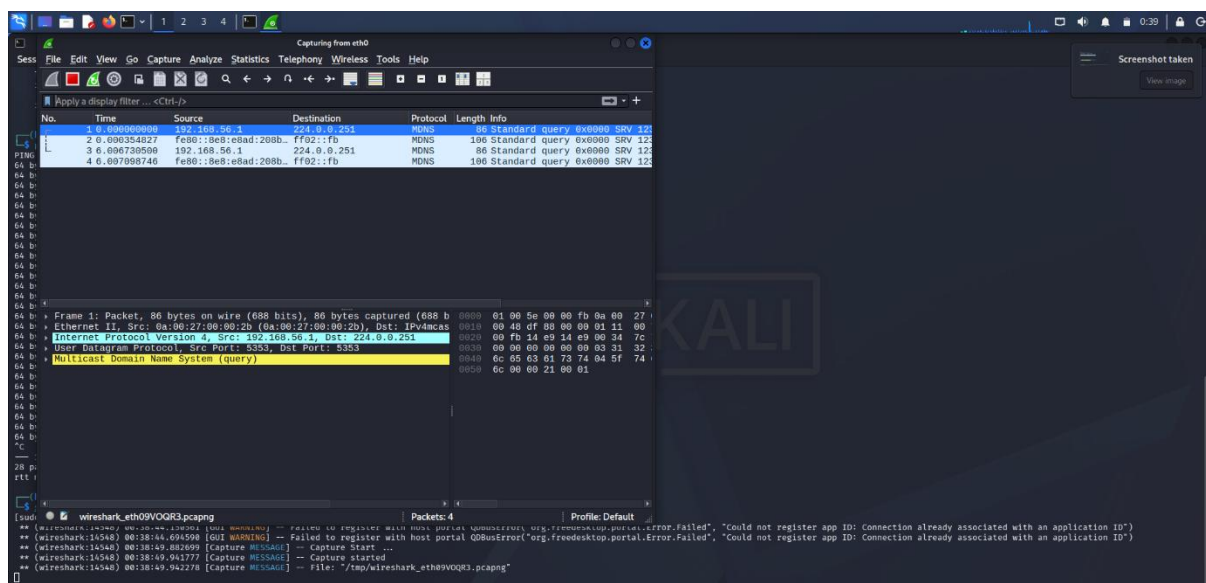
Wireshark was used to capture and analyze network packets exchanged between Kali Linux and Metasploitable2.

Procedure:

- Wireshark was launched in Kali Linux.
- The active network interface was selected.
- Packet capture was started.
- Network traffic was generated using the ping command.
- Captured packets were observed and analyzed.

Result:

Network packets were successfully captured and displayed within Wireshark.



7. PORT SCANNING USING NMAP

Nmap was used to identify open ports and active services running on Metasploitable2.

Command Used:

```
nmap 192.168.56.102
```

Sample Open Services Identified:

- FTP (Port 21)
- SSH (Port 22)

- Telnet (Port 23)
- HTTP (Port 80)

Result:

The scan successfully identified multiple open ports and services on the target machine.

```

kali@kali: ~
┌─── Session Actions Edit View Help
└─ ping 192.168.56.102 (192.168.56.102) 9(40) bytes of data:
  64 bytes from 192.168.56.102: icmp_seq=1 ttl=64 time=0.377 ms
  64 bytes from 192.168.56.102: icmp_seq=2 ttl=64 time=0.477 ms
  64 bytes from 192.168.56.102: icmp_seq=3 ttl=64 time=0.472 ms
  64 bytes from 192.168.56.102: icmp_seq=4 ttl=64 time=1.24 ms
  64 bytes from 192.168.56.102: icmp_seq=5 ttl=64 time=1.59 ms
  64 bytes from 192.168.56.102: icmp_seq=6 ttl=64 time=0.763 ms
  64 bytes from 192.168.56.102: icmp_seq=7 ttl=64 time=1.07 ms
  64 bytes from 192.168.56.102: icmp_seq=8 ttl=64 time=0.726 ms
  64 bytes from 192.168.56.102: icmp_seq=9 ttl=64 time=0.733 ms
  ^C
  --- 192.168.56.102 ping statistics ---
  9 packets transmitted, 9 received, 0% packet loss, time 8427ms
  rtt min/avg/max/mdev = 0.377/0.916/1.590/0.328 ms

kali@kali:~$ nmap 192.168.56.102
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-07 01:49 -0400
Nmap scan report for 192.168.56.102
Host is up (0.4002s latency).
Not shown: 677 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  setp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1899/tcp  open  rmiregistry
1924/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccpproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6080/tcp  open  x11
6667/tcp  open  irc
8080/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: 08:00:27:D9:EB:78 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 0.23 seconds

kali@kali:~$

```

8. CRYPTOGRAPHY USING OPENSSL

OpenSSL was used to demonstrate file encryption and decryption.

Procedure:

1. Created a text file named message.txt.
2. Added sample text content.
3. Encrypted the file using AES-256-CBC encryption.
4. Decrypted the encrypted file.
5. Verified the original content.

Commands Used:

```
echo "Hello ApexPlanet" > message.txt
```

```
openssl enc -aes-256-cbc -salt -in message.txt -out message.enc
```

```
openssl enc -d -aes-256-cbc -in message.enc -out decrypted.txt
```

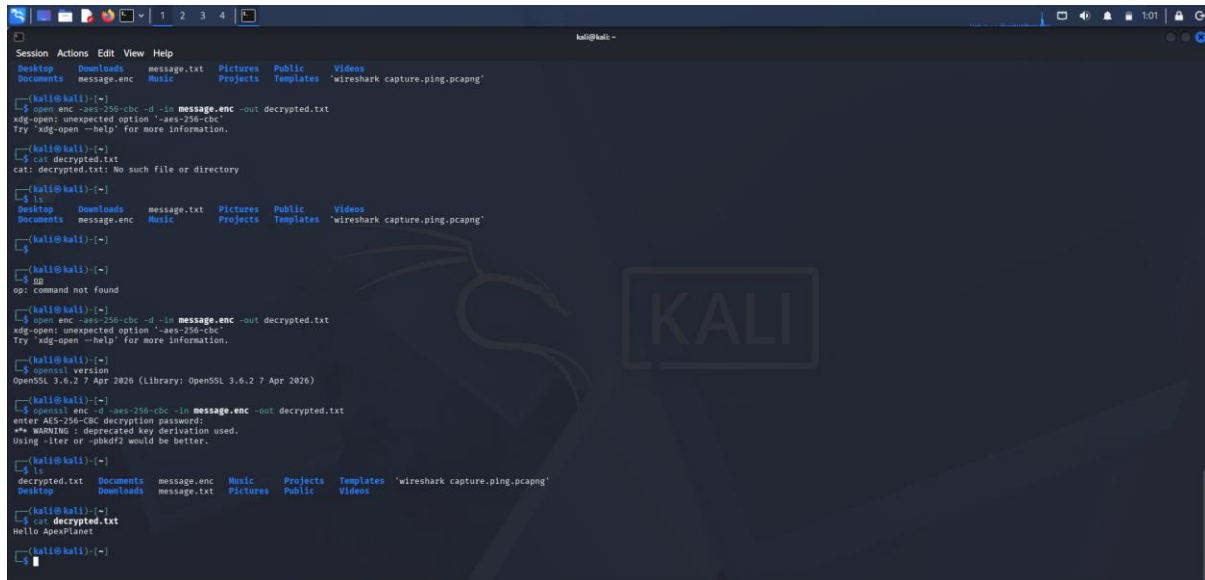
```
cat decrypted.txt
```

Output:

Hello ApexPlanet

Result:

The file was successfully encrypted and decrypted using OpenSSL.



```
kali@kali: ~  
$ openssl enc -aes-256-cbc -d -in message.enc -out decrypted.txt  
openssl: unexpected option "-aes-256-cbc"  
Try 'openssl --help' for more information.  
$ cat decrypted.txt  
cat: decrypted.txt: No such file or directory  
$ ls  
Desktop Downloads message.txt Pictures Public Videos  
Documents message.enc Music Projects Templates 'Wireshark capture.pcapng'  
$ cd  
op: Command not found  
$ openssl enc -aes-256-cbc -d -in message.enc -out decrypted.txt  
openssl: unexpected option "-aes-256-cbc"  
Try 'openssl --help' for more information.  
$ openssl version  
OpenSSL 3.0.2 7 Apr 2020 (Library: OpenSSL 3.0.2 7 Apr 2020)  
$ openssl enc -d -aes-256-cbc -in message.enc -out decrypted.txt  
enter AES-256-CBC decryption password:  
*** WARNING : deprecated key derivation used.  
Using -iter or -pbkdf2 would be better.  
$ ls  
decrypted.txt Desktop Downloads message.txt Music Projects Templates 'Wireshark capture.pcapng'  
$ cat decrypted.txt  
Hello ApexPlanet  
$
```

9. RESULTS

The following activities were completed successfully:

- VirtualBox installation and configuration
- Kali Linux setup
- Metasploitable2 setup
- Host-Only network configuration
- Connectivity verification using Ping
- Packet capture using Wireshark
- Port scanning using Nmap
- Encryption and decryption using OpenSSL

10. LEARNING OUTCOMES

Through this lab, the following concepts were learned:

- Virtual machine deployment
 - Network configuration in VirtualBox
 - Basic Linux command usage
 - Network troubleshooting
 - Packet analysis using Wireshark
 - Port scanning using Nmap
 - File encryption and decryption using OpenSSL
 - Practical cybersecurity laboratory setup
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11. CONCLUSION

The cybersecurity lab environment was successfully created using Kali Linux and Metasploitable2. Network communication between the virtual machines was established and verified. Wireshark was used to capture and analyze network traffic, Nmap was used to perform port scanning, and OpenSSL was used to demonstrate cryptographic operations. The lab provided practical experience with essential cybersecurity tools and techniques used in real-world security environments.